

Editor/Author Correspondence

Editor
2026-04-17
03:36 PM

Subject: [JMI ms#96541] Editorial Decision/Comments on "Healthcare Analytics Challenges: A Three-Pillar Framework Connecting Analytics Maturity, Workforce Agility, and Technical Enablement"

Delete

Decision E2: (not accepted for JMIR Medical Informatics)

Dear Samuel Harrold,

Thank you very much for your submission to JMIR Medical Informatics. The handling Editor has completed review of the article. At this time, the manuscript cannot be considered for publication in JMIR Medical Informatics.

Although this seems to be an interesting manuscript, the handling Editor deemed the work not suitable for consideration in JMIR Medical Informatics because of one or more of the following reasons:

- (X) The manuscript may be publishable in principle but was rated as "low impact".
- (X) The manuscript seems to have methodological weaknesses/limitations.

We should also note that JMIR Medical Informatics is trying to position itself as a leading eHealth journal, so we have to be selective in what we publish. If we do not accept your manuscript then this is not necessarily an indication of the scientific quality of your work, but sometimes more a reflection of our editorial priorities and interests.

Transfer the manuscript to another JMIR Publications journal and re-instate
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For more information about JMIR Publications transfer and reinstatement process, please read: <https://support.jmir.org/hc/en-us/articles/115001714547-How-do-I-request-a-manuscript-transfer-to-another-journal-if-my-paper-was-rejected-or-I-am-unable-to-address-reviewer-comments->

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Note that different JMIR Publications journals have different Article Processing Fees. Please check <https://jmir.org/fees/article-processing-fees> to verify the APF for the new journal.

Thank you very much again for choosing JMIR Medical Informatics and giving us the opportunity to review your interesting work. We realize that this may be disappointing news, but we hope that you will continue to consider JMIR Medical Informatics or another JMIR Publications journal as an outlet for your high-quality research work.

Best regards,

Anastasia Shelomentseva
anna.shelomentseva@jmir.org
Editorial Assistant

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[Ms link for editor only: <https://medinform.jmir.org/editor/submissionReview/96541>
For section/guest editor:
<https://medinform.jmir.org/sectionEditor/submissionReview/96541>]
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Questions? Check out our knowledge base at <https://support.jmir.org/hc/en-us> !
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JMIR Medical Informatics has a new Journal Impact Factor of 3.8 (Source: Journal Citation Reports 2025 from Clarivate); see <https://medinform.jmir.org/announcements/590>

JMIR Medical Informatics is indexed in PubMed, PubMed Central, DOAJ, SCOPUS, and SCIE (Clarivate)

JMIR Medical Informatics <http://medinform.jmir.org/>

Editor
2026-04-20
03:22 PM

Subject: [MxM ms#96541] Suggest Transfer for Your Recently Submitted Manuscript

[Delete](#)

Dear Samuel Harrold:

Thank you very much for your message.

After editorial review, your manuscript was deemed better suited for another JMIR Publications journal, namely one of the following journals:

- (X) Online Journal of Public Health Informatics
- (X) Interactive Journal of Medical Research (i-JMR)
- (X) JMIR Bioinformatics and Biotechnology **

**Please note that if you agree to transfer to JMIR Bioinformatics and Biotechnology and the Editor of the journal accepts the manuscript for publication after peer review, then JMIR Publications can offer a one-time APF waiver for the authors to use only for this journal and for this manuscript.

*Please check <https://jmir.org/fees/article-processing-fees> to verify the APF for the new journal.

Manuscript URL: <https://manuscripts.jmir.org/author/submission/96541>
Username: samuel.harrold@gmail.com

Please let us know within ONE (1) week whether or not you agree to a manuscript transfer and, if so, we will seek approval of the transfer from the Editor-in-Chief of the respective journal.

Important: If we do not hear from you within ONE (1) week regarding your preferences about the above options, we will release the manuscript back to you so you can decide on the best next steps for your work. The submission will be marked in our system as "Rejected with option to reinstate" to the suggested journal. Then, you may choose to reinstate your paper to one of the suggested journal(s).

Note that while we await your response about transfer, your manuscript is reassigned to the Transfer Hub.

For more information about JMIR Publications transfer process, please read: <https://support.jmir.org/hc/en-us/articles/115002462188-Why-has-my-article-been-transferred-to-another-journal-or-a-transfer-has-been-suggested-and-what-are-my-options>

Please let us know how you would like us to handle your submission by replying to this message or contacting ed-support@jmir.org.

Best regards,

Anastasia Shelomentseva
anna.shelomentseva@jmir.org
Editorial Assistant

[Ms link for editor only: <https://manuscripts.jmir.org/editor/submissionReview/96541>
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<https://manuscripts.jmir.org/sectionEditor/submissionReview/96541>]

Editor
2026-04-23
04:21 PM

Subject: Healthcare Analytics Challenges: A Three-Pillar Framework Connecting Analytics Maturity, Workforce Agility, and Technical Enablement

Samuel Harrold
Wed, Apr 22, 10:57 PM (17 hours ago)
to me, JMIR

Dear Anastasia Shelomentseva,

Thank you for your prompt response and for the transfer options provided.

I would like to proceed with a transfer to the Interactive Journal of Medical Research (i-JMR).

Please let me know if any additional information is needed to complete the transfer.

Best regards,
Samuel Harrold
samuel.harrold@yuimedi.com
Yuimedi, Inc.

Editor
2026-06-05
02:17 PM

Subject: [IJMR ms#96541] Editorial Decision/Comments on "Healthcare Analytics Challenges: A Three-Pillar Framework Connecting Analytics Maturity, Workforce Agility, and Technical Enablement" [Delete](#)

Decision D: (major revision & re-review)

Dear Samuel Harrold,

Your manuscript (#96541) in Interactive Journal of Medical Research has now been reviewed. The relevant reviewer/editor comments are listed at the end of this email.

Interactive Journal of Medical Research cannot offer publication of the paper in its current form. Please address the issues mentioned below and submit your revised manuscript for review. Substantial revisions may be noted in the comments.

Please contact us as soon as possible if you are unable to fully address the identified issues, or if you are requesting transfer to another JMIR Publications journal. See <https://jmir.zendesk.com/hc/en-us/articles/115001714547>

The due date for your revision and responses to reviewers is 2026-07-03.

This deadline can be extended once (see <https://jmir.zendesk.com/hc/en-us/articles/115001333167>).

Revision & Resubmission Instructions

To upload a revised version of your manuscript, go to <https://www.i-jmr.org/author/submissionReview/96541> and do the following:

- 1) Revised Manuscript: Use the "Revised Ms Upload" file-upload function and upload a clean (non-tracked) revised Word document of your manuscript.
- 2) Author Response to Comments: Copy and paste point-by-point responses to reviewer/editor comments into the notification email to the editor. Please do not upload a separate file containing responses. There are no formatting options (e.g., bold, italics) in this field.
- 3) Upload a revised version with tracked changes (or highlighted sections of text where changes were made) as a supplementary file, in addition to providing a clean version under "Revised Ms". See <https://jmir.zendesk.com/hc/en-us/articles/115001400448> for details on how to respond to reviewer comments and upload a revised manuscript. Supplying a version of the manuscript with tracked changes facilitates the review process.

Note that it is the authors' responsibility to ensure adherence to JMIR Publications style and formatting, including final word count.

Author names and metadata

All co-authors should verify how their names and affiliations will be displayed by informing the submitting author if edits are required.

Also note that for the final publication (if accepted), metadata (author names and affiliations) will be taken from the database information you entered when submitting this article, NOT the manuscript file.

Thank you.

Best regards,

Matthew Balcarras, PhD
matt.balcarras@jmir.org
(Editor)

Interactive Journal of Medical Research
JMIR Publications

For status updates, please first read the FAQ (<https://support.jmir.org/hc/en-us/articles/115001358188>).

For scientific content questions only, please use the "Contact Editor" function on your manuscript link (<https://support.jmir.org/hc/en-us/articles/115001332927>).

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[Ms link for editor only: <https://www.i-jmr.org/editor/submissionReview/96541>
For section/guest editor: <https://www.i-jmr.org/sectionEditor/submissionReview/96541>]

EDITORIAL COMMENTS (Decision: D/major revision)

Thank you for your submission. Please see the comments below and provide a point by point response.

* Editorial and content comments *

- The central argument of your viewpoint is interesting and relevant. However the manuscript could be stronger by using better examples and interacting more with relevant and recent literature.

* Formatting/structure comments *

- All in-text references must be numbers in square brackets. Do not use the author-year system, round brackets, or superscripts. Follow our instructions for authors for references: <https://support.jmir.org/hc/en-us/articles/115001333067-How-should-references-be-formatted-Which-journal-style-should-I-choose-when-using-Endnote->

- Do not number your headings or use italics or bold for emphasis in place of subheadings. Ensure each section has more than one subheading; otherwise, remove it. All subheadings must be under Introduction, Methods, Results, and Discussion.

- (required) Funding Statement - Add a funding statement, distinct from the Acknowledgements. If no funding was involved, please state: "The authors declared no financial support was received for this work." More information: <https://support.jmir.org/hc/en-us/articles/360015982471-How-should-the-Acknowledgments-section-be-formatted->

- (required) Disclose whether generative AI was used in any portion of the manuscript generation: if used, it is mandatory to include a written statement in the manuscript, even if you disclosed use at the time of submission (e.g., in cover letter or submission form). If none, authors should attest to this in the response to comments. <https://support.jmir.org/hc/en-us/articles/13387268671771-Do-you-allow-the-the-use-of-ChatGPT-or-other-generative-language-models-and-how-should-this-be-reported-> Authors may optionally use this tool to generate a complete disclosure: <https://panbibliotekar.github.io/gaidet-declaration/index.html>

Instructions

Please upload a revised version with tracked changes (or highlighted sections of text where changes were made) as a supplementary file, in addition to providing a clean version under "Revised Ms". See <https://jmir.zendesk.com/hc/en-us/articles/115001400448> for details on how to respond to reviewer comments and upload a revised manuscript. Supplying a version of the manuscript with tracked changes facilitates the review process.

Note that it is the authors' responsibility to ensure adherence to JMIR Publications style and formatting, including final word count.

Addressing formatting issues and any remaining comments during the revisions period speeds up the production process in the event of acceptance.

Further Editorial / Peer-reviewer Comments:

External Peer-Review Report(s)

Reviewer Q:
General comments
=====

This paper addresses an important issue regarding the internalization of tacit healthcare analytics knowledge within healthcare organizations which usually leaves the organization due to high turnover rate among experts. Authors recommend a human in the loop mechanism to translate the queries of users into implementable SQL queries and run them through experts for validation and potential correction. The validated queries are then stored in a knowledge base for future references. While the concern regarding knowledge restoration is a valid challenge across all organizations, including healthcare, the proposed framework has certain major issues which require further attention.

Specific comments
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Major comments

1. The manuscript frequently claims low maturity level by referring to certifications such as HIMSS. It is worth noting that organizations primarily avoid such assessments due to high costs. Therefore, concerns about self selection bias requires attention. In addition, using such acronyms without proper explanation adds to complexity of the manuscript.
 2. The practicality and efficiency of the proposed framework depends on the volume, diversity and frequency of the queries generated by users. A deeper discussion around these aspects of the the query generation activities within a healthcare organization is needed. For instance, if an organization requires only 50 frequent queries, then the problem can be addressed via a proper documentation or even the use of an industry level knowledge base.
 3. One of the biggest concerns which requires further investigation is the whether the human in the loop can actually perform better than AI or LLMs. an empirical investigation of the performance of experts in detecting and correcting AI errors in this context can significantly add to the value of the manuscript. If that fails, the effectiveness of the model will be a subject of debate.
 4. Finally, if the data is constantly changing, as has been argued in the manuscript, then the tenure of the informatics experts might not be enough to address the maturity issue. in addition, trainings and awareness will be key players too.
- ...

Minor comments

5. please make sure to clearly explain the acronyms used throughout the manuscript, otherwise it will be complex to follow the flow of the paper.
6. some of the reference, specially those arguing about the tenure of healthcare IT workers, is more than 20 years old and should be replaced.

Please help us improve the quality of reviewers by providing a rating for each review using the URLs provided below (your ratings are anonymous and only seen by the editor):

TO RATE REVIEWER Q: CLICK ON ONE OF THE FOLLOWING URLS

1. very poor/misleading/wrong statements DO NOT USE REVIEWER AGAIN:

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2. poor review (e.g. very superficial, extremely short):

[https://www.i-jmr.org/rev/authorRating?
ms=96541&re=603710&ro=1&ra=2&au=022cc970dc](https://www.i-jmr.org/rev/authorRating?ms=96541&re=603710&ro=1&ra=2&au=022cc970dc)

3. below average (too short or various weaknesses):

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4. average (solid review with minor weaknesses):

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ms=96541&re=603710&ro=1&ra=4&au=022cc970dc](https://www.i-jmr.org/rev/authorRating?ms=96541&re=603710&ro=1&ra=4&au=022cc970dc)

5. excellent / very useful review:

[https://www.i-jmr.org/rev/authorRating?
ms=96541&re=603710&ro=1&ra=5&au=022cc970dc](https://www.i-jmr.org/rev/authorRating?ms=96541&re=603710&ro=1&ra=5&au=022cc970dc)

Reviewer T:
General comments
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This viewpoint paper presents the author’s argument regarding persistent workforce turnover in the healthcare analytics domain and the associated challenges of retaining tacit knowledge. The author highlights that traditional apprenticeship models are increasingly ineffective, particularly given the short tenure of many professionals in healthcare settings. Although the issue itself is not novel, it remains a persistent and unresolved problem over time. To address this challenge, the author proposes leveraging artificial intelligence to capture and preserve tacit knowledge. The paper introduces a useful framework—human-in-the-loop knowledge governance (HITL-KG)—which aims to shift the locus of organizational knowledge from volatile human memory to durable semantic artifacts. These artifacts, termed “Validated Query Triples,” consist of a natural language intent, executable SQL, and accompanying rationale metadata, thereby enabling more sustainable knowledge retention and transfer.

Specific comments
=====

Major comments

Major Comments

1. Definition of Key Terms

Please clearly define the key terms used in the title—analytics maturity, workforce agility, and technical enablement. While their general meanings can be inferred, academic rigor requires that the author provide explicit definitions, ideally grounded in or adapted from existing literature. These are not novel constructs, and established definitions are likely available and should be cited to strengthen conceptual clarity.

2. Redundancy of Content

Some content appears repetitive. For example, statistics related to CIO tenure and turnover rates are introduced in the Introduction but are reiterated again in Section 2.1. Consider consolidating this information to avoid redundancy and improve the overall flow of the manuscript.

3. Clarification of Figure 1 (Framework Dynamics)

Figure 1 presents the architecture of the human-in-the-loop knowledge governance framework. However, the diagram currently depicts a static knowledge base and set of best practices. In reality, best practices are dynamic and evolve with new technologies and changing business rules. It would be more appropriate to reflect this iterative nature in the figure. Specifically, Step 7 (organizational memory) should feed back into Step 1 (knowledge base and best practices), following validation from prior steps, to illustrate continuous learning and system updating.

4. Use of Outdated Sources

Section 4.2 references a statistic from 2004. Given the rapidly evolving nature of healthcare analytics and workforce dynamics, this source is outdated. Please update with more recent evidence to ensure the argument reflects the current state of the field.

Minor Comment

• Please provide complete reference details for all citations, including volume, issue number, and page ranges. For example:
Ang, S., & Slaughter, S. (2004). Turnover of information technology professionals: The effects of internal labor market strategies. ACM SIGMIS Database: The DATABASE for Advances in Information Systems, [volume(issue)], pages.
<https://doi.org/10.1145/1017114.1017118>

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1. very poor/misleading/wrong statements DO NOT USE REVIEWER AGAIN:

<https://www.i-jmr.org/rev/authorRating?ms=96541&re=477063&ro=1&ra=1&au=ca9b670f90>

2. poor review (e.g. very superficial, extremely short):

<https://www.i-jmr.org/rev/authorRating?ms=96541&re=477063&ro=1&ra=2&au=ca9b670f90>

3. below average (too short or various weaknesses):

<https://www.i-jmr.org/rev/authorRating?ms=96541&re=477063&ro=1&ra=3&au=ca9b670f90>

4. average (solid review with minor weaknesses):

[https://www.i-jmr.org/rev/authorRating?
ms=96541&re=477063&ro=1&ra=4&au=ca9b670f90](https://www.i-jmr.org/rev/authorRating?ms=96541&re=477063&ro=1&ra=4&au=ca9b670f90)

5. excellent / very useful review:
[https://www.i-jmr.org/rev/authorRating?
ms=96541&re=477063&ro=1&ra=5&au=ca9b670f90](https://www.i-jmr.org/rev/authorRating?ms=96541&re=477063&ro=1&ra=5&au=ca9b670f90)

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Interactive Journal of Medical Research's Journal Impact Factor is 2.2 (Source:
Journal Citation Reports 2025 from Clarivate); see [https://www.i-
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Interactive Journal of Medical Research is indexed in Sherpa/Romeo, DOAJ, ESCI,
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Author
2026-06-15
03:42 PM

Subject: [IJMR ms#96541] Author Response/Comments on "Healthcare Analytics Challenges: A Three-Pillar Framework Connecting Analytics Maturity, Workforce Agility, and Technical Enablement" [Delete](#)

We thank the editor and both reviewers for their constructive and detailed comments. The revision strengthens the manuscript's engagement with recent peer-reviewed literature, adds explicit definitions of the three core constructs, addresses the scope and human-in-the-loop efficacy concerns with empirical evidence, and brings the formatting into compliance. In total we added 14 references spanning 1994-2026. Most are recent (2016-2026); the few older entries are foundational sources for the constructs and methods cited, each paired with a recent review: for example, Breu et al. 2002 (the origin of the workforce-agility construct) with Tessarini Junior & Saltorato 2021, and Tomaskovic-Devey et al. 1994 (foundational work on organizational survey non-response) with Halbesleben & Whitman 2013. Each comment is addressed below; section names refer to the revised manuscript.

EDITORIAL AND CONTENT COMMENTS

COMMENT (Editor): The central argument of your viewpoint is interesting and relevant. However the manuscript could be stronger by using better examples and interacting more with relevant and recent literature.

RESPONSE: We have substantially increased engagement with recent peer-reviewed literature throughout. Specifically, we added: cited scholarly definitions of the three title constructs (Reviewer T, comment 1); empirical evidence on the accuracy of human-in-the-loop oversight of AI-generated analytics (Reviewer Q, comment 3); database query-log evidence on query reuse and diversity (Reviewer Q, comment 2); methodological literature on self-selection and non-response bias in voluntary assessments (Reviewer Q, comment 1); and contemporary (2021-2025) workforce-turnover evidence. These additions are detailed in the responses below.

FORMATTING AND STRUCTURE COMMENTS (Editor)

COMMENT: All in-text references must be numbers in square brackets. Do not use the author-year system, round brackets, or superscripts.

RESPONSE: Done. The in-text citation style has been changed; references are now sequentially numbered and presented in square brackets (for example, [1], [2,3]), with no superscripts, author-year, or round brackets.

COMMENT: Do not number your headings or use italics or bold for emphasis in place of subheadings. Ensure each section has more than one subheading; otherwise, remove it. All subheadings must be under Introduction, Methods, Results, and Discussion.

RESPONSE: We have addressed each element:

- No headings are numbered.
- No section has exactly one subheading; each section has either no subheadings or two or more.
- The three short italic titles that previously labeled the sub-tables of the assessment rubric (Pillar 1, Pillar 2, Pillar 3) have been converted to proper descriptive subheadings, so italics are no longer used in place of subheadings. Remaining italics denote defined terms and list labels (emphasis), consistent with journal style.
- Regarding the requirement that all subheadings fall under Introduction, Methods, Results, and Discussion: this manuscript is a Viewpoint, and JMIR's own article-type instructions specify a different structure for Viewpoints than for Original Papers. Per "What are the article types for JMIR Publications journals?" (<https://support.jmir.org/hc/en-us/articles/115004950787-What-are-the-article-types-for-JMIR-Publications-journals>), the Viewpoint manuscript structure is: "Unstructured; using descriptive section headers that organize the work according to the content presented is strongly recommended. Headings such as 'Methods' and 'Results' are not allowed in Viewpoints." We have therefore used descriptive thematic headers throughout, as the Viewpoint article type requires, rather than IMRD headings. We would of course be glad to adjust if the editor prefers, recognizing that adopting an IMRD structure with Methods and Results headings would entail re-designating the article type away from Viewpoint.

COMMENT (required): Add a funding statement, distinct from the Acknowledgements.

RESPONSE: A Funding section is present and is distinct from the Acknowledgments. Because financial support was received, the "no financial support" statement does not apply; the Funding statement reads: "Yuimedi, Inc. provided financial support for

the author's time researching and writing this manuscript." This is consistent with the Conflicts of Interest disclosure of the author's role at Yuimedi.

COMMENT (required): Disclose whether generative AI was used in any portion of the manuscript generation.

RESPONSE: A generative AI disclosure is included in the manuscript (within Acknowledgments). It states that generative AI (Gemini CLI and Claude Code) was used to assist manuscript editing, language refinement, and literature search; that it was not used to generate research findings, data, or conclusions; and that the author conducted the research and verified all claims, citations, and AI-assisted text.

=====
REVIEWER Q
=====

COMMENT (Major 1): The manuscript frequently claims low maturity by referring to certifications such as HIMSS. Organizations primarily avoid such assessments due to high costs, so self-selection bias requires attention. In addition, acronyms are used without proper explanation.

RESPONSE: We added a self-selection-bias caveat in the Pillar 1 (Analytics Maturity) evidence: because HIMSS AMAM assessment is voluntary and self-selected, the reported figures likely over-represent analytics-committed organizations, so low maturity may be even more widespread than the counts suggest. This is supported by the organizational-survey methodology literature on non-response and self-selection bias [Halbesleben & Whitman 2013, Health Services Research; Tomaskovic-Devey et al. 1994, Administrative Science Quarterly]. We also broadened the maturity discussion beyond HIMSS by defining analytics maturity through the wider maturity-model literature (see Reviewer T, comment 1). Regarding cost as the specific mechanism of avoidance: we did not find an authoritative source attributing assessment avoidance specifically to cost, so we have framed the issue as self-selection bias grounded in the methodology literature rather than asserting an unsupported cost mechanism. On acronyms: all domain acronyms (HIMSS, AMAM, NL2SQL, EMR, EMRAM, LLM, ACO, EHR, CMS, CIO, DIKW, AACODS) are now defined at first use in the running text, and the Abbreviations list has been completed.

COMMENT (Major 2): The practicality of the framework depends on the volume, diversity, and frequency of queries. If an organization requires only 50 frequent queries, the problem can be addressed by documentation or an industry-level knowledge base.

RESPONSE: We added a scope-conditions discussion to the Comparison of Existing Approaches. Empirical query-log studies show that organizational analytical workloads are neither small nor static: more than half of analytical queries recur in large production workloads [Jindal et al. 2019], yet even small databases log thousands of distinct query strings [Kul et al. 2018]. A fixed set of documented queries therefore captures neither the large recurring core nor the evolving ad-hoc tail. HITL-KG addresses both regimes, reusing validated triples for recurring queries and capturing new ones at the point of use. This directly answers the "50 queries" objection: the recurring set is real but neither tiny nor static, and the diverse tail is exactly where documentation fails. The reviewer also raised the alternative of an industry-level knowledge base; the manuscript's Structural Barriers section addresses this, arguing that institution-specific business logic (local billing rules, coding conventions, and schema idiosyncrasies) is precisely what shared or industry-level knowledge bases cannot capture, as the failures of centralized efforts such as IBM Watson Health and Haven illustrate. HITL-KG captures this local logic rather than enforcing global standards.

COMMENT (Major 3): A central concern is whether the human-in-the-loop can actually perform better than AI or LLMs. An empirical investigation of experts detecting and correcting AI errors would significantly add value; otherwise the model's effectiveness is debatable.

RESPONSE: We agree this is the pivotal question. We have added empirical evidence to the Safety as Cognitive Forcing section showing that human oversight measurably improves AI-generated analytics: in a prospective study across six systematic reviews, large-language-model output verified by a human reviewer outperformed a human-only workflow (91.0% versus 89.0% accuracy) [Gartlehner et al. 2025, Annals of Internal Medicine]; in text-to-SQL specifically, an expert validation gate improved semantic correctness by 28.4% over end-to-end model output [Benzarti & Berrabah 2026]. We also cite balancing evidence that these gains are conditional on well-designed interaction [Ning et al. 2024]. Because this is a Viewpoint, we do not present an original empirical study; instead, we now frame HITL-KG explicitly as a prescriptive design whose comparative effectiveness is the subject of planned empirical validation, and the Implications section identifies the controlled evaluation (error rates, recovery time, and rework versus baseline tooling) as a priority for that

future work.

COMMENT (Major 4): If data is constantly changing, the tenure of informatics experts may not be enough to address the maturity issue; training and awareness will be key players too.

RESPONSE: We added a paragraph to Limitations acknowledging both points. HITL-KG complements rather than replaces workforce development: it decouples institutional knowledge from individual tenure but still presumes investment in training and analytics literacy [Konrad et al. 2022]. Moreover, precisely because retained expertise cannot manually keep pace with continual change in data models and business rules, knowledge must be externalized into curated artifacts; we also acknowledge that the validated query library itself requires active maintenance, which the Continuous Analytic Integration mechanism supports but does not eliminate.

COMMENT (Minor 5): Please clearly explain the acronyms used throughout the manuscript.

RESPONSE: Addressed as in comment 1: acronyms are now defined at first use, and the Abbreviations list is complete.

COMMENT (Minor 6): Some references, especially those on the tenure of healthcare IT workers, are more than 20 years old and should be replaced.

RESPONSE: The Workforce Agility evidence now relies on contemporary (2021-2025) data, including replacement-cost figures and current intent-to-leave and turnover statistics [for example, Rajamani et al. 2025; National Healthcare Retention report 2025]. The single remaining reference to the 2004 study (Ang & Slaughter) appears only once, in the Introduction, where it is explicitly framed as a two-decade-old benchmark whose continued use is itself evidence of the field's instability; the duplicate prior mention was removed.

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REVIEWER T
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COMMENT (Major 1): Please clearly define the key terms in the title (analytics maturity, workforce agility, technical enablement), ideally grounded in existing literature.

RESPONSE: We added explicit, cited definitions at the head of each evidence pillar. Analytics maturity is defined as an organization's progression in using data and quantitative models for fact-based decisions [Davenport & Harris 2007], operationalized through staged maturity models beyond the HIMSS model [Grossman 2018; Langer 2025]. Workforce agility is defined as a workforce's capacity for proactivity, adaptability, and resilience under change [Breu et al. 2002; Tessarini Junior & Saltorato 2021]. Technical enablement is defined as the technologies and practices that let non-specialist domain users access and analyze data without heavy reliance on central IT [Alpar & Schulz 2016; Bani-Hani et al. 2020].

COMMENT (Major 2): Some content is repetitive; for example, CIO tenure and turnover statistics are introduced in the Introduction and reiterated in Section 2.1. Consider consolidating.

RESPONSE: We removed the repeated statistics (the 53% CIO short-tenure figure and the 55% public-health-informatics intent-to-leave figure) from the "Broken Cycle" section, where they had duplicated the Introduction. These statistics are now stated once, at first mention in the Introduction, and the "Broken Cycle" section refers back to them while retaining only the figures unique to its argument.

COMMENT (Major 3): Figure 1 depicts a static knowledge base and best practices. Step 7 (organizational memory) should feed back into Step 1 (knowledge base and best practices) to illustrate continuous learning and system updating.

RESPONSE: We revised Figure 1 to add a feedback edge from Organizational Memory to the Knowledge Base, labeled "Curates best practices" (step 8). Validated query triples now update the ontologies and best practices that in turn provide context to future queries, closing a continuous-learning loop so that best practices evolve as new knowledge is validated rather than remaining static. The figure caption was updated to describe this loop.

COMMENT (Major 4): Section 4.2 references a statistic from 2004; please update with more recent evidence.

RESPONSE: Addressed together with Reviewer Q, comment 6. The 2004 statistic

was removed from that section, which now relies on contemporary (2021-2025) evidence; the only remaining mention is in the Introduction, explicitly framed as a historical benchmark.

COMMENT (Minor): Please provide complete reference details for all citations, including volume, issue number, and page ranges (for example, the Ang & Slaughter 2004 reference).

RESPONSE: The Ang & Slaughter (2004) reference is now complete: ACM SIGMIS Database, 35(3), 11-27, doi:10.1145/1017114.1017118. The newly added references were entered with full author lists, volume, issue, pages, and DOIs verified against Crossref. References that lack volume or page ranges are non-paginated sources (industry reports, institutional web resources, and press releases), cited with publisher and URL as appropriate.

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We thank the editor and reviewers again for their time. We believe the revisions materially strengthen the manuscript's evidentiary grounding, conceptual clarity, and formatting compliance, and we welcome any further guidance.

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[Link for editor: <https://www.i-jmr.org/editor/submissionReview/96541>
Link for section editor: <https://www.i-jmr.org/sectionEditor/submissionReview/96541>]

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Editor
2026-07-07
11:26 AM

Subject: [IJMR ms#96541] Editorial Decision/Comments on "Healthcare Analytics Challenges: A Three-Pillar Framework Connecting Analytics Maturity, Workforce Agility, and Technical Enablement" [Delete](#)

Decision B: (minor revision)

Dear Samuel Harrold,

Your manuscript (#96541) in Interactive Journal of Medical Research has now been reviewed. The relevant reviewer/editor comments are listed at the end of this email.

Please address the issues mentioned below and submit your revised manuscript for editorial review.

A minor revision decision is not a guarantee of a final decision of acceptance. Please note that the editor may determine upon re-review that the paper is better suited for publication in another JMIR Publications journal and recommend a manuscript transfer, especially if critical issues remain unaddressed or are introduced in the course of further revision.

The due date for your revision and responses to reviewers is 2026-07-14.

This deadline can be extended once (see <https://jmir.zendesk.com/hc/en-us/articles/115001333167>).

Revision & Resubmission Instructions

To upload a revised version of your manuscript, go to <https://www.i-jmr.org/author/submissionReview/96541> and do the following:

- 1) Revised Manuscript: Use the "Revised Ms Upload" file-upload function and upload a clean (non-tracked) revised Word document of your manuscript.
- 2) Author Response to Comments: Copy and paste point-by-point responses to reviewer/editor comments into the notification email to the editor. Please do not upload a separate file containing responses. There are no formatting options (e.g., bold, italics) in this field.
- 3) Upload a revised version with tracked changes (or highlighted sections of text where changes were made) as a supplementary file, in addition to providing a clean version under "Revised Ms". See <https://jmir.zendesk.com/hc/en-us/articles/115001400448> for details on how to respond to reviewer comments and upload a revised manuscript. Supplying a version of the manuscript with tracked changes facilitates the review process.

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All co-authors should verify how their names and affiliations will be displayed by informing the submitting author if edits are required.

Also note that for the final publication (if accepted), metadata (author names and affiliations) will be taken from the database information you entered when submitting this article, NOT the manuscript file.

Thank you.

Best regards,

Matthew Balcarras, PhD
matt.balcarras@jmir.org
(Editor)

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EDITORIAL COMMENTS (Decision: B)

Thank you for the revisions to your manuscript. Please note the comments below and provide a point by point response.

Further Editorial / Peer-reviewer Comments:

Please see this link for details on manuscript length, the current version is too long: https://support.jmir.org/hc/en-us/articles/115004950787-What-are-the-article-types-for-JMIR-Publications-journals#h_01GB60593SQ0DCKPYTARYGRV6S

The reference list includes items with incorrect or invalid DOIs. Please review all references for accuracy, appropriateness and validity.

Please ensure that all tables and figures are included in the manuscript in proper format so that they can be viewed correctly. The current version has text that should probably be in tables but is not formatted in a way that can be viewed clearly.

----- snip -----

Reviewer Q:
General comments
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Thank you for addressing the reviewers' comments. The current draft has sufficient potential for publication in IJMR.
My only recommendation is that the proposed model has strong potential to be further expanded and empirically evaluated. Two particularly promising directions for future work are:

1. Empirical assessment of the human-in-the-loop component. Specifically, the authors could investigate whether incorporating a human expert into the validation process leads to a measurable improvement in the accuracy and quality of the generated knowledge base compared to a fully automated approach.
2. Comparative evaluation against existing models. The proposed framework could be compared with established approaches in a controlled or simulated setting to assess differences in the quality, completeness, and reliability of the generated knowledge base when responding to the same set of inquiries.

While I do not consider these analyses essential for the current manuscript, I believe they represent valuable avenues for future research that would further strengthen and validate the proposed framework.

Please help us improve the quality of reviewers by providing a rating for each review using the URLs provided below (your ratings are anonymous and only seen by the editor):

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ms=96541&re=603710&ro=2&ra=3&au=a922e144bb

4. average (solid review with minor weaknesses):
[https://www.i-jmr.org/rev/authorRating?](https://www.i-jmr.org/rev/authorRating?ms=96541&re=603710&ro=2&ra=4&au=a922e144bb)
ms=96541&re=603710&ro=2&ra=4&au=a922e144bb

5. excellent / very useful review:
[https://www.i-jmr.org/rev/authorRating?](https://www.i-jmr.org/rev/authorRating?ms=96541&re=603710&ro=2&ra=5&au=a922e144bb)
ms=96541&re=603710&ro=2&ra=5&au=a922e144bb

Reviewer T:
General comments
Thank you for the revision. The author has satisfactorily addressed my concerns, and I have no further questions. One minor suggestion is to add figure captions. You currently have footnotes for each figure; please include captions as well.

Please help us improve the quality of reviewers by providing a rating for each review using the URLs provided below (your ratings are anonymous and only seen by the editor):

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2. poor review (e.g. very superficial, extremely short):
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3. below average (too short or various weaknesses):
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ms=96541&re=477063&ro=2&ra=3&au=b72fd30355

4. average (solid review with minor weaknesses):
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ms=96541&re=477063&ro=2&ra=4&au=b72fd30355

5. excellent / very useful review:
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